

CSPE-611(vi) (Advanced Machine Learning)

B.Tech 6th (NEP)

Time : 3 Hours

Max. Marks : 60

Note : The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all by selecting one question each from Section A, B, C and D. Section E is compulsory.

SECTION - A

1. (a) Define an Artificial Neural Network. Using suitable example(s), discuss the biological inspiration behind ANN and how the architecture of a biological neuron mimics a synthetic one. **(6)**
- (b) Explain the principle of MLE (Maximum Likelihood Estimation). How does it differ from Maximum A Posteriori (MAP) estimation? Explain using an example. **(6)**

OR

2. (a) Describe the structural differences between an MLP and a non-linear problem like XOR. Why is a hidden layer necessary for solving non-linear problems like XOR? Explain using a suitable example and/or diagram. **(6)**
- (b) Describe the complete Backpropagation algorithm. Explain how the 'Error Signal' propagates from the output layer back to the input layer, using a suitable example. **(6)**

SECTION - B

3. (a) Compare and contrast Sigmoid, Tanh and ReLU activation functions. What are the pros and cons of each? **(6)**
- (b) Explain the difference between Frequentist and Bayesian approaches to probability. How does Bayesian learning incorporate 'prior knowledge'? Explain using a suitable example. **(6)**

OR

4. (a) Explain the Naive Bayes algorithm for classification. Why is it 'computationally inexpensive'? Explain the issue that may result due to the conditional independence assumption, and why it may be a problem due to the conditional independence assumption. **(6)**
- (b) Explain the principle of MAP (Maximum A Posteriori) estimation? Explain using an example. **(6)**

SECTION - C

5. (a) Define Entropy in the context of information theory. Provide the formula and explain what it indicates about a dataset's value for attribute selection. **(6)**
- (b) Discuss why decision trees are highly susceptible to overfitting. Using an example, describe the different signs that indicate a decision tree has overfitted. **(6)**

OR

6. (a) Explain how a decision tree is represented as a tree. What do the internal nodes, branches, and leaf nodes signify? **(6)**
- (b) How do decision tree algorithms handle noise and missing attribute values in the training data? Analyze the time complexity for building a decision tree and for classifying a new instance. **(6)**

SECTION - D

7. (a) Define Reinforcement Learning. Explain the Agent-Environment interface and the concept of 'Exploration vs Exploitation' using an example scenario. Why is function approximation (like using a Neural Network) necessary for RL in high-dimensional state spaces? **(6)**
- (b) Explain the core philosophy of Ensemble Learning. Why does combining multiple 'Weak Learners' result in a 'Strong Learner'? **(6)**

OR

8. (a) Explain the concept of Boosting. Describe the sequential learning process where each new model corrects the errors of its predecessor. What type of weak learners are typically used in boosting? **(6)**
- (b) Using a suitable example, explain Random Forests in detail. Show how it combines Bagging with 'Feature Randomness'. **(6)**

SECTION - E

(Compulsory)

9. (a) What is the significance of the 'Bias' term in a Neural network layer?
- (b) What is the Symmetry Breaking problem?
- (c) What is the MAP Hypothesis?
- (d) What is a latent variable in the context of Bayesian learning?
- (e) Define Stop Criteria in the context of tree induction.
- (f) Define the Policy (π) in Reinforcement Learning. **(6×2=12)**

Transcribed from a photograph of the original answer/question booklet. A few numeric values, table entries and one section could not be fully verified due to image quality (angle/blur/overlap with adjacent papers) — please cross-check against your original copy.